Supplementary material: Adaptive certification of attack-conditional disclosure risk across release programmes

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References to theorems, propositions, assumptions, sections and tables without the prefix S are to the main paper.

S1 Filtration and conditional validity

The global filtration $(\mathcal{F}_g)_{g\geq 0}$ is generated by everything recorded up to global step g: development results, proposals, model updates, registrations, target draws and audit outcomes on every coordinate. Coordinate i is registered at the stopping time σ_i, and its bth outcome is recorded at $\gamma_{i,b}$, with $\mathcal{F}_{\gamma_{i,b}-} = \mathcal{F}_{\gamma_{i,b}-1}$ the σ-field just before that draw. Between its own draws the e-process of coordinate i is constant. At a draw, Assumption 2 requires the conditional success probability under the null to be at least τ_i given everything recorded so far, including outcomes on other coordinates. With the identity in the proof of Theorem 1, this makes the embedded process a nonnegative supermartingale in $(\mathcal{F}_g)$, that is, a test supermartingale [10, 9]. This condition suffices for arbitrary global monitoring; it is not necessary. Independence of trials within a coordinate is not enough if correlated results on another coordinate reveal a target's hidden type and so change the conditional success probability.

Relaxed denominator. Suppose an externally justified deviation $\kappa_i \geq 0$ satisfies $\mathbb{E}[X_{i,b} \mid \mathcal{F}_{\gamma_{i,b}-}] \geq \tau_i - \kappa_i$ under the null. Then replace τ_i in the denominators of (5) by $\tau_i - \kappa_i$, keep the policy tolerance τ_i in the null, and restrict the alternative to $q_{i,b} \in [0, \tau_i - \kappa_i)$. An alternative in $[\tau_i - \kappa_i, \tau_i)$ would make the expected factor exceed one for success probabilities just above $\tau_i - \kappa_i$. The change costs power. A drift statistic on features or a calibration metric does not by itself establish κ_i. Under cluster dependence, design trials around independent clusters or use an e-process valid for the cluster law; a reused target within a cluster is not a fresh audit unit.

Why marginal validity is not enough. Let $H \sim \text{Bernoulli}(1/2)$ be a hidden bit, drawn at registration and not observed, and set every attack result $X_b = H$ (the same target attacked repeatedly). Marginally $\mathbb{E}X_b = 1/2$, but after one failure the conditional probability of a later success is zero. An auditor who treats each repeat as a fresh Bernoulli(1/2) trial can drive the evidence as high as it likes by attacking the same failed target again. Assumption 2 fails, and no time-uniform error bound follows. Whether the assumption holds is a question of audit design, not of model calibration.

Weak and strict inequalities. The null $r_i \geq \tau_i$ includes the boundary, so a coordinate whose risk equals its tolerance is certified with probability at most its error share. An

agency that wants equality to count as safe can change the convention, but should not switch between strict and weak inequalities.

Proof details. On H_i, each factor has conditional expectation at most one given $\mathcal{F}_{\gamma_{i,b}-}$, by conditions (b), (c) and Assumption 2. Because (θ_i, a_i), τ_i and w_i are $\mathcal{F}_{\sigma_i}$-measurable and r_i is a function of the registered scenario, $\mathbf{1}_{H_i}$ is $\mathcal{F}_{\sigma_i}$-measurable; we set $w_i = 0$ on $\{\sigma_i = \infty\}$. The conditional form of Ville's maximal inequality for nonnegative supermartingales [12] gives $\mathbb{P}(A_i \mid \mathcal{F}_{\sigma_i})\mathbf{1}_{H_i} \le \alpha w_i \mathbf{1}_{H_i}$. Summing unconditional bounds over $i = 1, \dots, M$ and letting $M \to \infty$ by monotone convergence, the pathwise bound $\sum_i w_i \le 1$ gives the familywise bound. The argument holds when the rule for registering coordinate i depends on outcomes of coordinates $1, \dots, i-1$ and when audits are interleaved. It fails if $q_{i,b}$ is chosen after $X_{i,b}$ is seen, whatever it is later called.

S2 Allocation and zero-success calculation

The allocation $w_i = 6/(\pi^2 i^2)$ is fixed at the ith registration and sums to one over countably many registrations; it is an instance of online Bonferroni [11], the simplest case of alpha-spending [4]. An allocation that depends on the history at registration is also valid, provided the remaining budget is never overspent. The local threshold is $\log E_{i,b} \ge -\log(\alpha w_i)$, and each failure under a zero alternative adds $-\log(1-\tau_i)$. The implementation returns the trial at which the running sum first crosses the threshold, so it agrees with the gate in floating point: 29 trials for $\tau = 0.12$, $\alpha = 0.05$ and $w = 1/2$, and 40 for $w = 1/8$, where the threshold is $\log 160$.

Cost weights and integer effects. With planning drifts $\mu_i = \mu(p_i, q_i)$ from (8) and trial costs c_i for a frozen family, the expected-cost bound of Proposition 4 has the unique minimiser $w_i \propto c_i/\mu_i$; the zero-success case $p_i = q_i = 0$ gives $\mu_i = u_i = -\log(1-\tau_i)$, and u_i does not affect the minimiser. The rule gives more of the budget to costly audits and to coordinates planned close to their tolerance. It does not account for release utility, subgroup priority or scenarios added later, and a planning value that is too optimistic makes the audit longer, not invalid. Integer ceilings make the true objective a step function; neighbouring allocations can be enumerated after computing the continuous one, but must be fixed before outcomes are seen.

Reserve for later coordinates. Choose $0 < \beta < 1$. Give β to the known family by the cost rule and reserve $(1-\beta)\,6/(\pi^2 j^2)$ for the jth coordinate introduced later. The sequence is summable on every path, even when a later coordinate is triggered by earlier outcomes. A reserve is not protection against an unregistered harm.

S3 Numerical risk bounds and the audited margin

For a fixed sample size B and Y independent Bernoulli successes, the exact one-sided upper bound $\overline{R}$ solves $\sum_{m=0}^{Y} \binom{B}{m} \overline{R}^m (1-\overline{R})^{B-m} = \lambda$; at $Y = 0$, $\overline{R} = 1 - \lambda^{1/B}$ [1]. For a fixed family, use $\lambda = \alpha/M$ and a union bound [3]. If one inspects repeatedly and stops when the fixed-B bound is favourable, that coverage claim no longer holds. A time-uniform upper confidence

sequence, obtained by inverting a family of valid sequential tests indexed by hypothetical risks, can supply $\overline{R}_i$ for the audited margin [2, 7, 5, 6, 14] $z_i = 1 - \overline{R}_i/\tau_i$. The e-process at a single τ_i tests eligibility at that tolerance and does not output such a sequence.

A report should say, for every figure, whether it is a point estimate, a model prediction, an upper confidence bound or a tolerance. The crossing indicator of the gate belongs in a separate field with its level αw_i, the number of draws and the attack protocol. Reporting an e-process as a risk estimate is an error.

S4 A proposal model for bets

Any predictor may supply the bets $q_{i,b}$, provided each is fixed before $X_{i,b}$ is observed and clipped to $[0, \tau_i)$; validity does not depend on its accuracy (Section 4.5). The repository contains one such predictor, SENTRY-R, which fits a logistic regression or a small ReLU network to counted development attacks by the binomial log likelihood and ranks candidate releases by their largest predicted risk-to-tolerance ratio. It keeps three streams apart: scenarios proposed by a language model or a reviewer, which can enlarge the threat model but supply no outcomes; development attacks, which train the predictor; and final audit outcomes, which alone can certify. A risk probability asserted by a language model is not a Bernoulli outcome. On a synthetic programme law, over 10 seeds with both models trained to convergence, logistic regression had the lower test log loss in all 10 (0.2556 against 0.2576 for a three-layer network). This benchmark concerns SENTRY-R. The network of Section 5.3 is the neural release pipeline, `privacy_risk.neural_pipeline`; its development counts, fitted weights, frozen plans and final outcomes are in `artifacts-neural/neural_pipeline.json`. Training equations and the benchmark are in `docs/SENTRY_R.md`.

S5 Empirical methods and provenance

Carried-over results. The synthetic population of [8] has 200,000 units and a five per cent release sample; its four-release experiment draws 20,000 targets and intersects the candidate sets of four 3-anonymous key projections. The synthetic credit register has six monthly vintages; for each policy the linkage evaluation uses 4,000 attacks on independently drawn targets, with one exposure reading per vintage corrupted by Gaussian noise. Three harms are evaluated beyond record identification: precision of an alleged lender relationship, disclosure of arrears status, and differencing. The main paper uses identification only. The public Census sample holds 29,501 records; neither it nor the other public file used in [8] supplies the population from which it was sampled. The release and pooled columns of Table 5 and the grid used to choose the minimum cell count come from separate target draws.

Toy audit. `scripts/adaptive_audit_comparison.py` uses seed 2709, 2,000 replications per law, at most 300 draws, $\alpha = 0.05$, $w = 1/8$, $\tau = 0.12$, $q = 0.03$ and true risks 0.04 and 0.12. The first-passage event is $\max_{b \le 300} \log E_b \ge \log 160$. The fixed comparator uses only the count at draw 300 and its exact one-sided bound at level αw. In the adaptive run a new stream with risk 0.12 is created only if an independent stream with risk 0.04 has crossed. Monte Carlo standard errors in Table 1 are $\sqrt{\hat{p}(1-\hat{p})/2000}$.

Register. The synthetic register of `src/privacy_risk/credit.py` draws 20,000 borrowers, 60 lenders and 32,581 loans over six months. A share of loans is originated after the first month, each loan leaves with a monthly hazard of 0.04 and does not return, and the stock stays near 27,000. This replaces the panel of [8], in which every loan was forced to be live in the first month and a third of loans left in the second month and came back in the third, which gave the history adversary an artificial signal. Exposure is lognormal within size class and constant across vintages.

Credit-register audit. `scripts/adaptive_audit_chain.py` runs the release chain over the floors 300 to 1 with the three declared attacks, the bet $\min\{(Y+1/2)/(B+1), 0.11\}$, at most 3,000 fresh targets per coordinate and threshold $1/\alpha$, using `AdaptiveAudit.register_release_chain`. The five other designs of Section 5.2 are applied to the same streams. True risks come from separate draws of 100,000 targets, on seeds disjoint from the audit seeds. The operating-characteristics check runs the release chain 400 times on Bernoulli streams at the estimated risks, and at risks raised by 0.01 and 0.02.

Factor-by-factor floors. `scripts/programme_robustness.py` chooses the floor for ten adversaries (the five of Table 6, the insider with log-exposure error 0.01, and four with exposure published to one significant figure) by fixed-sequence testing on one draw of 4,000 targets per level and confirms it on a second draw; where confirmation fails, the next stricter level that passes is used, which keeps the error bound because every stricter level also passed at selection. It audits the largest size class on its own stratified draw, counts the loans whose key and published exposure are unique (the relink share under rotating pseudonyms), records the share of loans, exposure, large-firm loans and multi-lender borrowers kept at each floor, gives floors at tolerances of 0.05, 0.09, 0.12 and 0.20, and repeats the choice over 20 register draws and reading standard deviations of 0.3, 0.6 and 1.0 with 2,000 targets per level. At the declared noise of 0.6 the floors over register draws were 1 for one release, 40 to 75 (median 75) for pooling with the history, and 100 to 150 (median 150) for the full attack; at 0.3 they were 20, 150 and 300, with no level passing in 3 of 20 draws; at 1.0 they were 1, 40 and 75. `scripts/robustness_table.py` writes the tables.

Audit-process attacks. `scripts/audit_process_attacks.py` simulates 20,000 Bernoulli streams per risk with one coordinate at weight one, and 2,000 runs, each of at most 300 draws, of an operator that bets just below the tolerance after each success and zero after each failure. The rates for ten and a hundred tries are computed as $1-(1-f)^{\text{tries}}$ from the single-try rate f, not simulated. The controls it models are in `src/privacy_risk/adaptive_audit.py`: `commit` and `target_seed` derive targets from a committed secret and a beacon round, `CommittedCoordinate` records each bet in an `AuditLedger` before the attack runs, and the ledger is hash-chained. The script simulates the effect of these controls on Bernoulli streams; it does not call them.

S6 Algorithm and reproduction

1. Fix the harm taxonomy and the pooled attack protocol, and keep the release history and data versions.

2. Register each coordinate with its scenario, tolerance and weight w_i, drawn from the remaining budget. Registered fields cannot be changed.

3. Before each attack choose $q_{i,b} \in [0, \tau_i)$ from earlier information only. Record the target draw, outcome, bet, sampling-unit identifier and attack code digest.

4. Add $\log(q/\tau)$ to the log evidence on a success and $\log((1-q)/(1-\tau))$ on a failure; a success with $q = 0$ sets it to $-\infty$. Each step adds the log likelihood ratio of Bernoulli(q) to Bernoulli(τ), as in the sequential probability ratio test [13].

5. Certify at the first crossing of $-\log(\alpha w_i)$ and accept no further outcomes. Call a release eligible only when every declared coordinate is certified, and record the owner and the expiry of the decision.

The following commands, run from the repository root, install the repository, test the gate, recreate Tables 1, 2, 6, 7 and 8, and rerun the pipeline of [8] that produces the carried-over results:

```
python -m pip install -e '.[dev]'
python -m pytest tests/test_adaptive_audit.py -q -o addopts=""
python scripts/adaptive_audit_comparison.py
  -output paper/publication_versions/adaptive_audit_toy.json
python scripts/adaptive_audit_chain.py
python scripts/programme_robustness.py
python scripts/robustness_table.py
python scripts/audit_process_attacks.py
python -m privacy_risk.pipeline -config configs/paper.json
  -output artifacts-paper
```

The code checks inputs and the error budget; it cannot check that attack draws satisfy Assumption 2 or that the declared coordinate set is complete.

S7 Relation to earlier work by the author

Table 5 and the four-release and public-file figures in Section 5.4 of the main paper were first reported in [8], an author manuscript whose earlier version was presented at the Banco de Portugal and IFC Workshop on Data Security and Privacy, Lisbon, September 2026. That study did not run the gate. Theorem 1, Propositions 2 to 6, the toy audit, the credit-register audit on the revised register, the factor-by-factor floors and the audit-process measurements are new to the present article.

S8 Multiscale proposal models and audited distribution shift

This extension separates three mathematical objects: the unknown attack law, the fitted proposal function, and a confidence bound from new attack trials. Its results do not replace the registered sequential gate. In particular, a derivative of a neural network describes its response to an input change; it does not bound an adversary's true success probability. The new software is `privacy_risk.multiscale`. All new numerical demonstrations use synthetic laws and do not modify the earlier public-data results.

S8.1 An arbitrary dimensional risk vector

Let $x \in \mathcal{X} = \prod_{j=1}^{d}[\ell_j, u_j]$ describe a release and its registered threat model, with $u_j > \ell_j$. For coordinate $i \in \{1, \dots, m\}$, let $A_i(Z, x) \in \{0, 1\}$ be the declared attack success event. Define

$$r_i(x; P_i) = \int A_i(z, x)\, dP_i(z), \qquad z_i(x) = 1 - r_i(x; P_i)/\tau_i, \quad 0 < \tau_i < 1. \tag{S1}$$

A coordinate includes its target population, information available to the attacker, attack algorithm and harm. These components cannot change silently when the feature vector changes. Eligibility is the intersection of all registered inequalities $r_i \le \tau_i$, rather than an average over harms. The scalar $\min_i z_i$ records the limiting margin, but a report retains the whole vector so that a favourable component cannot conceal another violation.

S8.2 Finite functional wavelet dictionary

Normalize $v_j = (x_j - \ell_j)/(u_j - \ell_j)$ and define

$$\psi(t) = (1 - t^2)e^{-t^2/2}, \qquad \psi'(t) = (t^3 - 3t)e^{-t^2/2}. \tag{S2}$$

For levels $a = 0, \dots, J-1$ and centres $c_{ak} = (k + 1/2)2^{-a}$, $k = 0, \dots, 2^a - 1$, the feature map is

$$\Phi_J(x) = \big(v_j, \psi(2^a(v_j - c_{ak}))\big)_{j,a,k} \in \mathbb{R}^D, \qquad D = d\, 2^J. \tag{S3}$$

The dictionary captures localized changes at several scales and retains the normalized identity coordinates. It is neither an orthogonal wavelet transform nor an exact interpolant. On a bounded interval the atoms need not be orthogonal. An exact interpolation claim would require specified nodes, an invertible collocation matrix and a conditioning estimate, none of which follows from using a wavelet-shaped activation. Tensor products of all atoms cost exponentially in d; the coordinatewise dictionary avoids that construction. Cross-feature interactions arise in the dense network. Its computational benefit is a choice of representation, not a universal approximation-error guarantee.

Put $s_j = (u_j - \ell_j)^{-1}$. Each feature depends on a single raw input, so the feature Jacobian has orthogonal columns. Since $|\psi'| \le C_\psi = 3\sqrt{3}e^{-3/2} + 3e^{-1/2}$,

$$\|D\Phi_J(x)\|_2 \le L_\Phi = \max_j s_j \left(1 + C_\psi^2 \sum_{a=0}^{J-1} 2^a 4^a\right)^{1/2}. \tag{S4}$$

The bound follows by separately maximizing $|t|^3 e^{-t^2/2}$ and $3|t|e^{-t^2/2}$. Integration on a line segment in the convex box establishes $\|\Phi_J(x) - \Phi_J(y)\|_2 \le L_\Phi \|x - y\|_2$. Increasing resolution increases L_Φ and may worsen robustness; fine scales are not automatically an improvement.

S8.3 Deep counted likelihood and explicit sensitivities

For column-vector notation let $h_0 = \Phi_J(x)$, $h_l = \mathrm{ReLU}(W_l h_{l-1} + b_l)$, $l = 1, \dots, L$, and $f_\theta(x) = \sigma(W_{L+1} h_L + b_{L+1}) \in (0, 1)^m$. For development counts k_{ni} out of b_{ni} trials, fit

$$\mathcal{L}(\theta) = -\frac{1}{\sum_{n,i} b_{ni}} \sum_{n,i} \{k_{ni} \log f_{ni} + (b_{ni} - k_{ni}) \log(1 - f_{ni})\} + \frac{\lambda}{2} \sum_l \|W_l\|_F^2. \tag{S5}$$

Binomial counts retain different trial denominators. Treating all observed fractions as equally precise generally changes this objective. If the counts are dependent, the expression is a working loss, not an independent binomial likelihood. Training, early-stopping validation and final audits must use separate groups; neither LLM proposals nor final certification trials provide training labels.

Away from ReLU kinks the raw-input derivative is

$$Df_\theta(x) = D_\sigma W_{L+1} D_L W_L \cdots D_1 W_1 D\Phi_J(x), \tag{S6}$$

where D_l contains activation indicators and $D_\sigma = \operatorname{diag}(f_i(1-f_i))$. The implementation stores transposed weights because observations are rows. At ReLU kinks it uses the zero subgradient. For logits clipped to $[-35, 35]$, it additionally sets the logit derivative to zero outside that interval. Thus it differentiates the actual deployed predictor, rather than an unclipped surrogate.

Because ReLU and clipping are one-Lipschitz and $\sigma' \le 1/4$,

$$\|f_\theta(x) - f_\theta(y)\|_2 \le L_f \|x - y\|_2, \quad L_f = \tfrac{1}{4} L_\Phi \prod_{l=1}^{L+1} \|W_l\|_2. \tag{S7}$$

This global bound is conservative; local Jacobians can be much smaller. For coordinate i, the proposal margin derivative is $D\widehat{z}_i = -Df_i/\tau_i$. A useful sensitivity report gives raw units, feature ranges, point derivatives and the spectral bound. It never labels $\widehat{z}_i$ an audited margin. If a separate uniform approximation guarantee $\sup_x |r_i(x) - f_i(x)| \le \epsilon_i$ were established, the triangle inequality would imply $r_i(y) \le f_i(x) + L_f\|y - x\| + \epsilon_i$. A held-out mean log loss does not establish that uniform guarantee.

S8.4 A finite-family audit with a stated shift radius

Fix the family, sample sizes and weights before final outcomes are observed. For coordinate i, let $K_i \sim \operatorname{Bin}(n_i, p_i)$ under audit law P_i, and let Q_i be the intended deployment attack law. Assume an externally justified total-variation bound

$$\operatorname{TV}(P_i, Q_i) = \sup_E |P_i(E) - Q_i(E)| \le \rho_i. \tag{S8}$$

For weights $w_i > 0$, $\sum_i w_i \le 1$, use the exact one-sided Clopper–Pearson bound

$$U_i = \begin{cases} \operatorname{Beta}^{-1}(1 - \alpha w_i; K_i + 1, n_i - K_i), & K_i < n_i, \\ 1, & K_i = n_i. \end{cases} \quad U_i^Q = \min\{1, U_i + \rho_i\}. \tag{S9}$$

With probability at least $1 - \alpha$, all deployment risks obey $r_i(Q_i) \le U_i^Q$. To prove this, each binomial bound fails with probability at most αw_i; the union bound controls their simultaneous failure. On the complementary event, the definition of total variation applied to the attack-success event gives $r_i(Q_i) \le p_i + \rho_i \le U_i + \rho_i$. No independence across coordinates is needed. Independent Bernoulli trials within each coordinate, a fixed sample size and a fixed target law are needed. Random data-dependent radii require their own simultaneous error budget. A feature displacement and a model Lipschitz bound do not justify a TV radius.

The shifted audited position is $z_i^Q = 1 - U_i^Q/\tau_i$; all must be nonnegative. This is a conditional statistical guarantee given the external shift model, not proof that the radius is true or that unregistered harms are absent. At zero successes, $U_i = 1 - (\alpha w_i)^{1/n_i}$, so passage requires

$$n_i \geq \left\lceil \frac{\log(\alpha w_i)}{\log(1 - \tau_i + \rho_i)} \right\rceil, \qquad 0 \leq \rho_i < \tau_i. \tag{S10}$$

If $\rho_i \geq \tau_i$, zero observed successes cannot supply a finite passing budget. Repeatedly inspecting these fixed-size bounds and stopping at passage invalidates their stated coverage; use the existing sequential gate instead.

S8.5 Audit cost and scaling

For positive costs c_i, set $a_i = c_i/[-\log(1 - \tau_i + \rho_i)]$. Ignoring integer ceilings, minimize $C(w) = \sum_i a_i \log(1/(\alpha w_i))$ over positive weights summing to one. The Lagrangian derivative is $-a_i/w_i + \nu = 0$; summing gives

$$w_i^* = a_i / \sum_j a_j. \tag{S11}$$

The Hessian is diagonal with entries $a_i/w_i^2 > 0$, proving a unique minimum. The integer plan rounds upward and checks the exact zero-success inequality. It is a planning rule conditional on zero successes, not a promised audit cost under positive risk and not an optimizer of the discontinuous integer objective. Freeze weights from costs and declared tolerances before audit outcomes.

Feature construction costs $O(Nd2^J)$. A dense network forward pass costs $O(N \sum_l d_{l-1} d_l)$; storing every raw-input sensitivity additionally costs $O(Nmd)$ for the output alone. The implementation supports arbitrary d but limits J to eight to avoid accidental dictionary growth. Dimension-free statistical safety does not imply dimension-free computational or predictive accuracy. LLM scenario injections continue through the existing review and measured-development-evidence pipeline; modifying a dictionary or threat law requires a new proposal-model version and new final audits.

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